

No.	Solution
1 (a)	The ohm is defined as being the resistance of a conductor through which a current of one ampere is flowing when the potential difference across it is one volt.
1(b)(i)	$\rho = \frac{RA}{l}$ $= \frac{\left(\frac{1.50}{0.32}\right) \frac{\pi(0.23 \times 10^{-3})^2}{4}}{40.0 \times 10^{-2}} = 4.87 \times 10^{-7} \Omega \text{ m}$
1(b)(ii)	$\frac{\Delta \rho}{\rho} = \frac{\Delta V}{V} + \frac{\Delta I}{I} + 2 \frac{\Delta d}{d} + \frac{\Delta l}{l}$ $= \frac{0.01}{1.50} + \frac{0.01}{0.32} + 2 \frac{0.01}{0.23} + \frac{0.1}{40.0}$ $= 0.127$ $\Delta \rho = 6.20 \times 10^{-8} \Omega \text{ m} = 0.6 \times 10^{-7} \Omega \text{ m}$
1(b)(iii)	$\rho = (4.9 \pm 0.6) \times 10^{-7} \Omega \text{ m}$ A common error was to state the uncertainty to more than one significant figure.
1(c)	Accuracy refers to the closeness of the measured value, $\rho = (4.9 \pm 0.6) \times 10^{-7} \Omega \text{ m}$, to the actual value of $\rho = 4.4 \times 10^{-7} \Omega \text{ m}$. In this case, the measured value is accurate within the uncertainties of the quantities measured Precision refers to the degree of spread of the value obtained. For the measured value, the percentage spread is about 12% and thus it is not precise.
2 (a)	From Newton's 2 nd law,